# YOUSSEF NADA

**Computer Engineering Student | Backend (Java, Spring Boot) & AI/ML**
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## PROFESSIONAL SUMMARY

Computer and Control Systems Engineering student (expected 2028) who builds backend and AI-powered software. Hands-on experience with Java, Spring Boot, REST APIs, SQL databases (MySQL, PostgreSQL), Python machine learning, and Flutter. Built a real-time chat application with WebSocket and STOMP, a traffic prediction system with 93% accuracy, and a published real estate mobile app. Seeking an internship or junior role in backend, AI, or software engineering.

## EDUCATION

**Bachelor of Engineering in Computer and Control Systems Engineering**
*Mansoura University, Egypt* | *Expected graduation: 2028*
**Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Programming, Embedded Systems, Machine Learning, Computer Vision, Artificial Intelligence

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, Dart, C++, SQL
**Backend & Databases:** Spring Boot, Spring Data JPA, REST APIs, WebSocket, STOMP, MySQL, PostgreSQL, Firebase, Cloudinary
**AI / Machine Learning:** Machine Learning, Computer Vision, Deep Learning (basics), Scikit-learn, Pandas, NumPy, Matplotlib, Streamlit
**Mobile Development:** Flutter
**Core Concepts:** Object-Oriented Programming, Data Structures, Algorithms, Layered Architecture
**Tools:** Git, GitHub, VS Code, Google Colab
**Soft Skills:** Problem Solving, Teamwork, Communication, Fast Learning

## EXPERIENCE

**AI Intern | SYNTECXHUB | 2026**
- Completed hands-on training in Artificial Intelligence and Machine Learning workflows.
- Worked on data preprocessing, model training, evaluation, and deployment pipelines.
- Applied practical AI concepts in real-world projects using Python and ML libraries.

**Backend and Mobile Developer | Self-Directed Projects**
- Built backend applications with Java and Spring Boot, including a CRUD REST API with a layered architecture and relational databases (MySQL, PostgreSQL).
- Built and deployed cross-platform mobile applications using Flutter, integrating backend services, authentication, and APIs.
- Focused on clean architecture, performance optimization, and responsive, modern UI/UX.

## PROJECTS

**Smart Traffic Management System | Machine Learning | 2026**
*Python, Scikit-learn, Pandas, NumPy, Matplotlib, Streamlit* |
https://github.com/Youssefnada-Engineer/smart-traffic-management
- Built an end-to-end AI-powered system to predict traffic congestion and optimize signal timing.
- Generated and analyzed a realistic dataset of 34,560 records across 16 intersections over 90 days.
- Compared Random Forest, Gradient Boosting, and Decision Tree models, reaching 93% accuracy; tuned hyperparameters with GridSearchCV.
- Developed an adaptive signal control simulation that reduced waiting queues by 83%.
- Deployed an interactive Streamlit web app with a Traffic Dashboard, Congestion Predictor, and Signal Control Simulator.

**Real-Time Chat Application | Backend Development**
*Java, Spring Boot, WebSocket, STOMP*
- Implemented real-time, bidirectional messaging using WebSocket with the STOMP protocol for structured message exchange.
- Built the Spring Boot backend that handles messaging, delivering messages instantly without repeated HTTP polling.

**Nada City | Real Estate Mobile Application (Published)**
*Flutter, REST APIs | Available on Google Play and the Apple App Store*
- Developed and published a real estate mobile application for property listings (sale and rent) and finishing service inquiries.
- Designed a responsive UI for property browsing and search, integrated backend services and REST APIs, and managed the production release.

## CERTIFICATIONS

Flutter Development Course - Tharwat Samy | Machine Learning Course - Mansoura University | Computer Vision Course - NTI | Prompt Engineering - One Million Experts (Dubai) | Introduction to Artificial Intelligence - Microsoft | Introduction to Artificial Intelligence - IBM | Getting Started with Deep Learning - NVIDIA